

with his or her own RNN instance (with a common initial state shared among all speakers), and the system keeps updating the RNN state given the new embeddings from this speaker.

Representing speakers as RNN states enables the system to learn the high-level knowledge shared across different speakers and utterances using RNN parameters. This promises more labelled data. In contrast, common clustering algorithms almost always work with each single utterance independently, making it difficult to benefit from a large amount of labelled data.

Google plans to refine the model so that it can integrate contextual information to perform offline decoding, with further reduced DER. The tech giant is also planning to add some acoustic features to train entire speaker diarisation systems in an end-to-end way.

Develop, test and deploy intelligent robotics applications with AWS RoboMaker

Amazon Web Services Inc. (AWS) has announced the availability of AWS RoboMaker, a new service that makes it easy to develop, test and deploy robotics applications at scale.

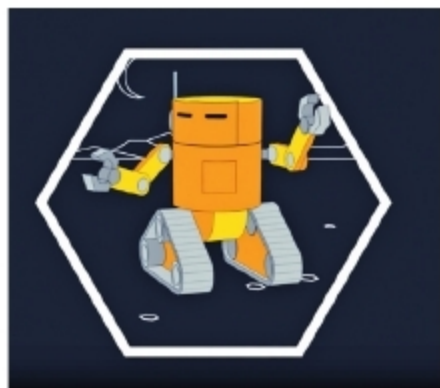
AWS RoboMaker provides four core capabilities—cloud extensions for Robot Operating System (ROS), a development environment, simulation and fleet management. ROS is the most widely used open source robotics software framework.

RoboMaker provides extensions for cloud services like Amazon Kinesis (video streaming), Amazon Rekognition (image and video analysis), Amazon Lex (speech recognition), Amazon Polly (speech generation), and Amazon CloudWatch (logging and monitoring) to developers who are using ROS.

These services are exposed as ROS packages, making it easier to use them to build intelligent functions into robotics applications without having to learn a new framework or programming language.

AWS RoboMaker provides an AWS Cloud9-based robotics integrated development environment for building and editing robotics applications. Its development environment includes the operating system, development software, and ROS automatically downloaded, compiled, and configured. Plus, RoboMaker cloud extensions and sample robotics applications are pre-integrated in the environment, so developers can get started in minutes.

AWS RoboMaker provides a fully managed robotics simulation service that supports large scale and parallel simulations, and automatically scales



the underlying infrastructure based on the complexity of the simulation. It also provides pre-built virtual 3D worlds such as indoor rooms, retail stores, and race tracks.

AWS RoboMaker provides a fleet management service that has robot registry, security, and fault-tolerance built-in so that developers can carry out over-the-air (OTA) updates and manage robotics applications

throughout the life cycle of the robots. It includes sample robotics applications to help developers get started quickly. These provide the starting point for the voice command, apart from recognition, monitoring, and fleet management capabilities that are typically required for intelligent robotics applications.

Because AWS RoboMaker is pre-integrated with popular AWS analytics, machine learning and monitoring services, it's easy to add functions like video

Increasing ICT expenditure by governments a boon for global open source LMS market

The global open source learning management system (LMS) software market is expected to witness significant growth in the forthcoming years. The growth, according to a new study, will be driven by the rise in information and communications technology (ICT) expenditure by governments of various developed and developing regions, including North America, Europe and the Asia-Pacific.

Analysts at Research N Reports have projected the global LMS market to grow at a compound annual growth rate (CAGR) of +9.26 per cent till 2025.

The Learning Management System (LMS) is a software application for managing, recording, tracking, reporting and providing educational courses or training programmes, or learning and development programmes. Although the first LMS appeared in the higher education sector, most LMSs today focus on the corporate market.

Prominent LMS solution providers, such as Litmos Limited, Oracle Corporation, Paradiso Solutions, Saba Software Inc., etc, offer LMS using cloud and on-premise deployment models.

According to the study, the Asia-Pacific LMS market is likely to register the highest CAGR in the near future. This is because countries such as China, India, Japan and Australia are offering lucrative opportunities for market players.

During the research, the analysts also observed that global vendors are investing a lot of capital in the market to improve the overall LMS solution offering, to ensure low cost and optimised employee training modules and solutions. They noted that an increase in demand for low cost cloud deployment and the surge in 'bring your own devices' (BYOD) are key drivers of the European LMS market.

Titled 'Global Open Source Learning Management Systems Software—Market Size, Status and Forecast 2025', the research report evaluates the historical and the current performance of this market.